



fixes_result.ipynb
File



notebooke36f51db96.ipynb
File

explain each and everything
result code meaning every code block
and the summary too
the theory the concepts the approach

Alright — this is a **serious research-level notebook**, so I'll walk you through it **end-to-end** in a structured way:

I'll cover:

1. Big Picture (What this project is doing)
2. Core Theory (Concepts used)
3. Each Major Code Block Explained
4. Training + Evaluation Logic
5. Results + Why things behaved like this
6. Final Summary (what you should say to professor)



1. BIG PICTURE (WHAT THIS PROJECT IS)

This is a **binary classification** problem:

- 👉 Detect Exoplanets (1) vs Non-Exoplanets (0)
- 👉 Input = time-series data (stellar brightness / flux)

Problem Type:

- Time series classification
- Highly imbalanced dataset (~150:1)



Core Challenge

- Very few exoplanets

- Models tend to predict only "0" (majority class)
- Accuracy becomes **misleading**

2. CORE THEORY (VERY IMPORTANT)

◆ (A) Class Imbalance Handling

Code:

 Python



 Run

Meaning:

- Assign higher importance to rare class

 Example:

- Class 0 $\rightarrow$ weight = 0.5
- Class 1 $\rightarrow$ weight = 75

 Model gets penalized more for missing exoplanets

◆ (B) FOCAL LOSS

 Python



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Why?

Normal BCE fails in imbalance.

Focal Loss Idea:

- Focus more on **hard samples**
- Ignore easy ones

Case

Behavior

Easy sample

Down-weighted

Case	Behavior
Hard sample	High penalty

👉 This is VERY IMPORTANT for your viva

◆ (C) Evaluation Metrics

From this function:

</> Python

Run

Metrics used:

Metric	Meaning
ROC-AUC	Ability to separate classes
PR-AUC	Better for imbalance
F1-score	Balance of precision & recall
Precision	How many predicted positives are correct
Recall	How many actual positives detected
MCC	Best overall metric for imbalance

Confusion Matrix:

TN FP
FN TP

Term	Meaning
TP	Correct exoplanet

Term	Meaning
FP	False alarm
FN	Missed exoplanet
TN	Correct non-exoplanet



3. MODEL ARCHITECTURES

◆ (1) BiLSTM

</> Python



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Concept:

- Reads sequence:
 - Forward →
 - Backward ←

Why useful?

Transit dip depends on context before & after

◆ (2) Multi-Channel BiLSTM

```
input_flux  
input_fft
```

</> Python



▶ Run

Idea:

Combine 3 different views:

Channel	Meaning
Raw flux	Original signal
FFT	Frequency patterns
Transit features	Physics-based features

👉 Then merged:

Python



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🔥 This is **feature fusion** (very important concept)

◆ (3) CNN Models

- CNN-1D → works on sequence
- CNN-2D → works on spectrogram

Concept:

- Detect patterns like:
 - dips
 - periodic shapes

◆ (4) FNN (Fully Connected)

- Basic baseline model
- Uses extracted features

◆ (5) ESN (Echo State Network) ★

Python



Run

Concept:

- Reservoir computing
- Random recurrent network (fixed weights)

👉 Only output layer is trained

Key Parameters:

Parameter	Meaning
spectral_radius	controls stability
leaking_rate	memory decay
sparsity	% of zero connections

⚙️ 4. DATA PROCESSING & AUGMENTATION

◆ Stratified Sampling

</> Python



▶ Run

👉 Keeps class ratio same in splits

◆ Data Augmentation

```
np.roll()  
noise
```

</> Python



▶ Run

What it does:

Technique	Purpose
Shift	simulate time delay
Noise	robustness
Scaling	brightness variation

👉 Applied **ONLY** to minority class (very important)

5. TRAINING PIPELINE

◆ K-Fold Cross Validation

 Python



 Run

Why?

- More reliable results
- Avoids lucky/unlucky splits

◆ Trials

 Python



 Run

👉 Multiple runs with different randomness

◆ Model Registry

 Python



 Run

👉 Stores:

- model
- training data
- test data

6. SANITY CHECK

</> Python



▶ Run

Purpose:

- Run quick test before full training
- Check:
 - no NaN
 - model works

7. CHECKPOINTING

</> Python



▶ Run

👉 Saves progress → avoids retraining

8. THRESHOLD CALIBRATION (VERY IMPORTANT)

Problem:

Default:

</> Python



Run

👉 Model predicts all zeros

Solution:

(1) Youden's J

</> Python



Run

(2) F1-max threshold

👉 Choose threshold giving best F1

🔥 THIS FIXES YOUR "F1 = 0" ISSUE

9. VISUALIZATIONS

ROC Curve

- X-axis: False Positive Rate
- Y-axis: True Positive Rate

👉 Higher curve = better

PR Curve

- Precision vs Recall

👉 More important for imbalance



10. RESULTS ANALYSIS

From your data:

ESN ROC-AUC = 0.826 (BEST)



Observation:

Model	Performance
ESN	Best
Multi BiLSTM	Good
CNN	Medium
GRU	Poor



WHY ESN WON?

From markdown:

Reasons:

1. Time-series memory captured better
2. No overfitting
3. Efficient representation



11. LIMITATIONS

- ESN not fully tuned
- Deep models need more data
- Noise still present



12. FINAL SUMMARY (WHAT YOU SHOULD SAY)



Use this in viva / PPT:

Problem

Detect exoplanets from highly imbalanced time-series data.

Approach

- Applied preprocessing + augmentation
 - Used multiple models:
 - BiLSTM, CNN, GRU, FNN, ESN
 - Used **Focal Loss + Class Weights**
 - Performed **Stratified K-Fold validation**
 - Applied **threshold optimization**
-

Key Insight

Accuracy is misleading → used:

- ROC-AUC
 - PR-AUC
 - F1-score
-

Result

- ESN achieved best ROC-AUC (~0.826)
 - Deep learning models underperformed
-

Conclusion

Reservoir Computing is highly effective for:

- small data
 - time-series
 - imbalanced problems
-

If you want next step

I can:

- Explain each line of a specific cell
- Help you defend this in front of professor
- Or convert this into a perfect PPT / viva script

Just tell me 👍



.+

Looks like you sent just ".+" — not sure what you want next 😊

Here are a few directions you can choose from (just reply with the number or tell me):

What do you want now?

1. Line-by-line explanation of every code cell (very detailed, like teaching a beginner)
2. Only the important/high-weight parts for viva (what prof will ask)
3. Fixes + improvements in your notebook (exact changes "replace this with this")
4. Convert this into PPT content
5. Explain only one model (like BiLSTM / ESN) deeply
6. Explain results section + how to justify them strongly

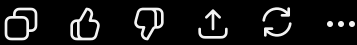
Right now I've already given you the full overview —
next step depends on how deep you want to go.



6

Great — this is the most important part for scoring marks.
I'll help you explain + defend your results like a pro.





Is this conversation helpful so far?

